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# Text-to-Image Generation using a Pre-trained Diffusion Model

!pip install diffusers transformers accelerate torch --quiet

import torch
from diffusers import StableDiffusionPipeline

# Load pre-trained Stable Diffusion model
model_id = "runwayml/stable-diffusion-v1-5"

# Note: To avoid the deprecation warning in newer diffusers versions,
# you can pass torch_dtype directly inside the dictionary or load it
cleanly.
pipe = StableDiffusionPipeline.from_pretrained(
    model_id,
    torch_dtype=torch.float32
)

# Use GPU if available
device = "cuda" if torch.cuda.is_available() else "cpu"
pipe = pipe.to(device)

# Take prompt from user
prompt = input("Enter your image prompt: ")

# Generate image
image = pipe(prompt).images[0]

# Display image
display(image)

# Save generated image
image.save("generated_image.png")

print("Image generated and saved as generated_image.png")

/usr/local/lib/python3.13/dist-packages/diffusers/utils/
deprecation_utils.py:23: FutureWarning: `torch_dtype` is deprecated
and will be removed in version 1.0.0. Please use `dtype` instead.
    deprecate("torch_dtype", "1.0.0", _TORCH_DTYPE_DEPRECATION_MESSAGE)

{"model_id": "120fd30ecdfb474abe634f95a04819ed", "version_major": 2, "vers
ion_minor": 0}

{"model_id": "a8999a5f94734bbaa03bb31ddebc02eb", "version_major": 2, "vers
ion_minor": 0}

{"model_id": "1ec23eb243dc452ab52e029c58a18c06", "version_major": 2, "vers
ion_minor": 0}

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{"model_id":"66fbdd2ef0f641e08f211261df24eb4d","version_major":2,"version_minor":0}
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

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{"model_id":"b8c908aa2afa4405886fab0ebfa051de","version_major":2,"version_minor":0}
```

```
{"model_id":"6d5d3381b10644aabebe963f5972ecc2","version_major":2,"version_minor":0}
```

```
{"model_id":"3c148b9bf69b49a3946357bfe05d2928","version_major":2,"version_minor":0}
```

Enter your image prompt: A beautiful sunset over mountains, digital art

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{"model_id":"3f717703001548f2bf466cc1f57b9a5e","version_major":2,"version_minor":0}
```



Image generated and saved as generated_image.png